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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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1. Introduction

The volume of oil produced into the storage tank at standard conditions is smaller than the volume of oil that flows from the reservoir to the bottom of the producing well. This change in oil volume is due to three factors:

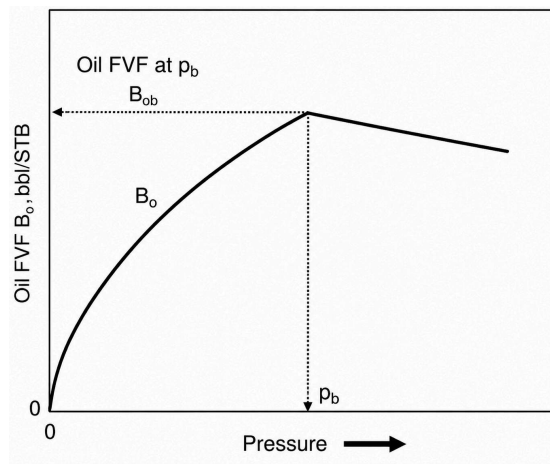
- Release of gas dissolved in the oil as pressure decreases from reservoir pressure to surface pressure.
- The reduction in pressure causes a slight expansion of the remaining oil.
- The remaining oil also contracts due to the decrease in temperature.

The oil formation volume factor, B_o , is defined as the volume of reservoir oil required to produce one stock tank barrel of oil at atmospheric conditions. The reservoir oil volume includes the gas dissolved in the oil.

$$B_o = \frac{\text{volume of oil at reservoir conditions}}{\text{volume of oil at stock-tank conditions}}$$

Oil volumes at surface or stock-tank conditions are conventionally reported at 60°F. Similarly, both stock-tank liquid volumes and surface gas volumes are expressed at standard conditions. The oil formation volume factor, B_o , is defined as the reservoir volume occupied by one stock tank barrel (STB) of oil, including the gas dissolved in solution, at reservoir pressure and temperature.

The following figure shows the typical variation of the oil formation volume factor with reservoir pressure for a black-oil reservoir fluid.



There are several numerical correlations available to compute the oil formation volume factor, B_o , some of the most widely used include:

- Standing
- Glaso
- Petrosky

2. Numerical solution

$p < p_b$

$B_o = 0.9759 + 12 \times 10^{-5} F^{1.2}$ $F = R_{sb} \sqrt{\frac{\gamma_g}{\gamma_o} + 1.25T}$

$p > p_b$

$B_o = B_{ob} \times \exp[c_o(p_b - p)]$

2.1 Import libraries

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [4]: gamma_g=0.9
gamma_o=31
T=np.array([100,110,120])

pb=2200
co=5*10**-6

p=np.linspace(100,3000,100)
```

2.3 Calculation and plotting

Note: To calculate the oil formation volume factor B_o is required to know the solution gas-oil ratio R_s . Please review the corresponding notebook

In [5]:

```
plt.figure(dpi=300, figsize=(8,4))

for i in range(len(T)):

    # Properties at bubble point
    Rsb = gamma_g*((pb/18.2 + 1.4)*10**((0.0125*gamma_o - 0.00091*T[i]))**1.2048
    Fb = Rsb*(gamma_g/gamma_o)**0.5 + 1.25*T[i]
    Bob = 0.9759 + 12e-5*Fb**1.2

    Bo_values = []

    for j in range(len(p)):

        # Bo at pressure Lower than pb
        if p[j] < pb:
            Rs = gamma_g*((p[j]/18.2 + 1.4)*10**((0.0125*gamma_o - 0.00091*T[i]))**1.2048

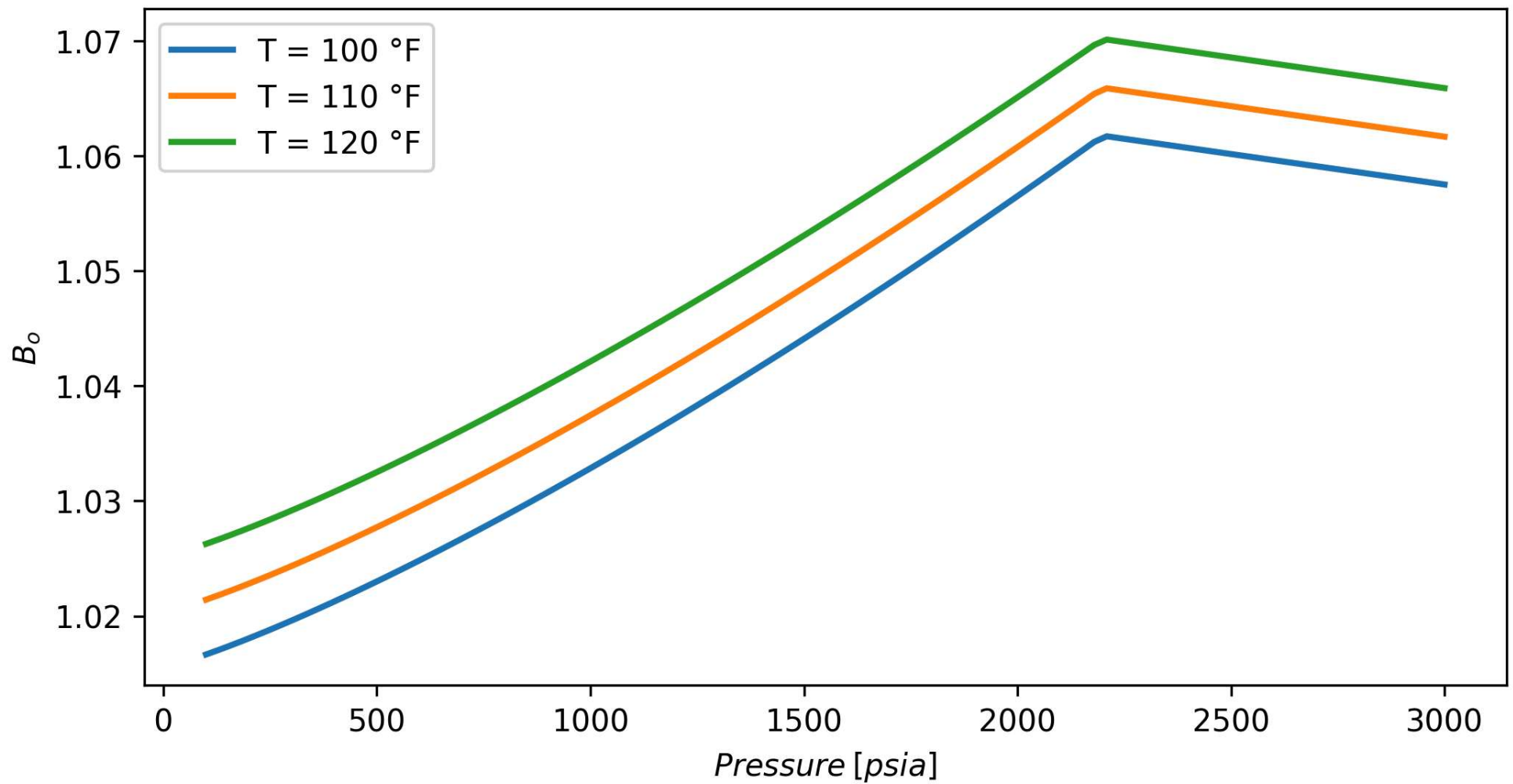
            F = Rs*(gamma_g/gamma_o)**0.5 + 1.25*T[i]
            Bo = 0.9759 + 12e-5*F**1.2

        # Bo at pressure greater than pb
        else:
            Bo = Bob*np.exp(co*(pb - p[j]))

        Bo_values.append(Bo)

    plt.plot(p, Bo_values, linestyle='-', linewidth=2,
             label=f'T = {T[i]} °F')

plt.xlabel('$Pressure\:[psia]$')
plt.ylabel('$B_o$')
# plt.grid(True)
plt.legend()
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: